

Analytical Distance Calculations between von Mises and Wrapped Cauchy Distributions

Naoki Otani

June 22, 2026

1 Introduction

This document summarizes the mathematical details of the analytical distance calculations (KL divergence and 1-Wasserstein distance) between the von Mises distribution and the wrapped Cauchy distribution on the circle S^1 .

2 Distributions

Let P be the von Mises distribution with mean direction $\mu_P \in [-\pi, \pi)$ and concentration parameter $\kappa > 0$:

$$p(\theta) = \frac{1}{2\pi I_0(\kappa)} e^{\kappa \cos(\theta - \mu_P)} \quad (1)$$

where $I_n(\kappa)$ is the modified Bessel function of the first kind of order n .

Let Q be the wrapped Cauchy distribution with mean direction $\mu_Q \in [-\pi, \pi)$ and concentration parameter $\rho \in [0, 1)$:

$$q(\theta) = \frac{1 - \rho^2}{2\pi(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))} \quad (2)$$

3 Kullback-Leibler Divergence

3.1 Trigonometric Moments Preparation and Derivations

For circular probability distributions symmetric about their mean direction μ , the trigonometric moments with respect to the centered angle $\theta - \mu$ are real-valued. Specifically, the sine moments vanish due to symmetry, while the cosine moments represent the characteristic function values.

3.1.1 Trigonometric Moments of the von Mises Distribution

For the von Mises distribution P with mean direction μ_P and concentration parameter $\kappa > 0$, the moments are defined as:

$$E_P[\cos(n(\theta - \mu_P))] = \frac{1}{2\pi I_0(\kappa)} \int_{-\pi}^{\pi} \cos(n(\theta - \mu_P)) e^{\kappa \cos(\theta - \mu_P)} d\theta \quad (3)$$

$$E_P[\sin(n(\theta - \mu_P))] = \frac{1}{2\pi I_0(\kappa)} \int_{-\pi}^{\pi} \sin(n(\theta - \mu_P)) e^{\kappa \cos(\theta - \mu_P)} d\theta \quad (4)$$

Let $\phi = \theta - \mu_P$. Under this change of variables, the domain of integration remains $[-\pi, \pi]$ due to the periodicity of the integrand. For the sine moment, the integrand $\sin(n\phi) e^{\kappa \cos \phi}$ is an odd function of ϕ , which implies:

$$E_P[\sin(n(\theta - \mu_P))] = 0 \quad (5)$$

For the cosine moment, we recall the standard integral representation of the modified Bessel function of the first kind of integer order n :

$$I_n(\kappa) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{\kappa \cos \phi} \cos(n\phi) d\phi \quad (6)$$

Substituting this definition into the expression for the expectation yields:

$$E_P[\cos(n(\theta - \mu_P))] = \frac{I_n(\kappa)}{I_0(\kappa)} \quad (7)$$

3.1.2 Trigonometric Moments of the Wrapped Cauchy Distribution

For the wrapped Cauchy distribution Q with mean direction μ_Q and concentration parameter $\rho \in [0, 1)$, we evaluate the complex trigonometric moments using contour integration. By definition:

$$E_Q[e^{in(\theta - \mu_Q)}] = \int_{-\pi}^{\pi} e^{in(\theta - \mu_Q)} \frac{1 - \rho^2}{2\pi(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))} d\theta \quad (8)$$

Letting $\phi = \theta - \mu_Q$, and using the Euler identity $2 \cos \phi = e^{i\phi} + e^{-i\phi}$, we can factor the denominator as:

$$1 + \rho^2 - 2\rho \cos \phi = (1 - \rho e^{i\phi})(1 - \rho e^{-i\phi}) \quad (9)$$

This gives:

$$E_Q[e^{in(\theta - \mu_Q)}] = \frac{1 - \rho^2}{2\pi} \int_{-\pi}^{\pi} \frac{e^{in\phi}}{(1 - \rho e^{i\phi})(1 - \rho e^{-i\phi})} d\phi \quad (10)$$

Let $z = e^{i\phi}$ be a complex variable on the unit circle C in the complex plane, so $dz = ie^{i\phi} d\phi = iz d\phi$, which implies $d\phi = \frac{dz}{iz}$. Since $e^{-i\phi} = 1/z$,

we rewrite the integral as a contour integral over the unit circle C oriented counterclockwise:

$$\begin{aligned} E_Q[e^{in(\theta-\mu_Q)}] &= \frac{1-\rho^2}{2\pi} \oint_C \frac{z^n}{(1-\rho z)(1-\rho/z)} \frac{dz}{iz} \\ &= \frac{1-\rho^2}{2\pi i} \oint_C \frac{z^n}{(z-\rho)(1-\rho z)} dz \end{aligned} \quad (11)$$

The integrand $f(z) = \frac{z^n}{(z-\rho)(1-\rho z)}$ has poles at $z = \rho$ and $z = 1/\rho$. Since $\rho \in [0, 1)$, only the simple pole $z = \rho$ lies inside the unit circle C . By the Residue Theorem, for $n \geq 0$:

$$\oint_C f(z) dz = 2\pi i \cdot \text{Res}(f, \rho) \quad (12)$$

Calculating the residue at $z = \rho$:

$$\text{Res}(f, \rho) = \lim_{z \rightarrow \rho} (z - \rho) \frac{z^n}{(z - \rho)(1 - \rho z)} = \frac{\rho^n}{1 - \rho^2} \quad (13)$$

Substituting this back, we obtain:

$$E_Q[e^{in(\theta-\mu_Q)}] = \frac{1-\rho^2}{2\pi i} \left(2\pi i \frac{\rho^n}{1-\rho^2} \right) = \rho^n \quad (14)$$

Taking the real and imaginary parts of $E_Q[e^{in(\theta-\mu_Q)}] = \rho^n$, we get:

$$E_Q[\cos(n(\theta - \mu_Q))] = \rho^n \quad (15)$$

$$E_Q[\sin(n(\theta - \mu_Q))] = 0 \quad (16)$$

3.2 Fourier Expansion of the Log-Cosine Term

To evaluate expectations involving the log-denominator of the Wrapped Cauchy density, we require the Fourier series expansion of $\log(1 + \rho^2 - 2\rho \cos x)$.

3.2.1 Derivation of the Expansion

Let $z = \rho e^{ix}$ be a complex number. Since $\rho \in [0, 1)$, we have $|z| = \rho < 1$. We use the Taylor series expansion of the complex logarithm $\log(1 - z)$ for $|z| < 1$:

$$\log(1 - z) = - \sum_{n=1}^{\infty} \frac{z^n}{n} = - \sum_{n=1}^{\infty} \frac{\rho^n}{n} e^{inx} \quad (17)$$

Taking the complex conjugate, we also have:

$$\log(1 - \bar{z}) = - \sum_{n=1}^{\infty} \frac{\bar{z}^n}{n} = - \sum_{n=1}^{\infty} \frac{\rho^n}{n} e^{-inx} \quad (18)$$

Adding these two identities together yields:

$$\begin{aligned}\log(1 - z) + \log(1 - \bar{z}) &= -\sum_{n=1}^{\infty} \frac{\rho^n}{n} (e^{inx} + e^{-inx}) \\ &= -2 \sum_{n=1}^{\infty} \frac{\rho^n}{n} \cos(nx)\end{aligned}\quad (19)$$

Using the properties of the logarithm, the left-hand side can be combined as:

$$\begin{aligned}\log(1 - z) + \log(1 - \bar{z}) &= \log((1 - z)(1 - \bar{z})) \\ &= \log(1 - z - \bar{z} + z\bar{z}) \\ &= \log(1 - \rho(e^{ix} + e^{-ix}) + \rho^2) \\ &= \log(1 + \rho^2 - 2\rho \cos x)\end{aligned}\quad (20)$$

Equating the two expressions, we obtain the desired Fourier series representation:

$$\log(1 + \rho^2 - 2\rho \cos x) = -2 \sum_{n=1}^{\infty} \frac{\rho^n}{n} \cos(nx) \quad (21)$$

3.3 Divergence from von Mises to Wrapped Cauchy

The Kullback-Leibler (KL) divergence $D_{\text{KL}}(P \parallel Q)$ from P to Q is defined as:

$$D_{\text{KL}}(P \parallel Q) = E_P \left[\log \frac{p(\theta)}{q(\theta)} \right] = E_P[\log p(\theta)] - E_P[\log q(\theta)] \quad (22)$$

First, we evaluate $E_P[\log p(\theta)]$. Taking the logarithm of the density of the von Mises distribution:

$$\log p(\theta) = -\log(2\pi) - \log I_0(\kappa) + \kappa \cos(\theta - \mu_P) \quad (23)$$

Taking the expectation with respect to P and using the prepared moment $E_P[\cos(\theta - \mu_P)] = \frac{I_1(\kappa)}{I_0(\kappa)}$, we obtain:

$$\begin{aligned}E_P[\log p(\theta)] &= -\log(2\pi) - \log I_0(\kappa) + \kappa E_P[\cos(\theta - \mu_P)] \\ &= -\log(2\pi) - \log I_0(\kappa) + \kappa \frac{I_1(\kappa)}{I_0(\kappa)}\end{aligned}\quad (24)$$

Next, we evaluate $E_P[\log q(\theta)]$. The logarithm of the wrapped Cauchy density is:

$$\log q(\theta) = \log(1 - \rho^2) - \log(2\pi) - \log(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q)) \quad (25)$$

To calculate the expectation of the last term, we use the Fourier series expansion of the logarithm:

$$\log(1 + \rho^2 - 2\rho \cos x) = -2 \sum_{n=1}^{\infty} \frac{\rho^n}{n} \cos(nx) \quad (26)$$

Setting $x = \theta - \mu_Q$, we have:

$$E_P[\log(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))] = -2 \sum_{n=1}^{\infty} \frac{\rho^n}{n} E_P[\cos(n(\theta - \mu_Q))] \quad (27)$$

To evaluate $E_P[\cos(n(\theta - \mu_Q))]$ when the mean directions are not aligned, we decompose the angle as $\theta - \mu_Q = (\theta - \mu_P) + (\mu_P - \mu_Q)$ and apply the trigonometric addition formula:

$$\begin{aligned} \cos(n(\theta - \mu_Q)) &= \cos(n(\theta - \mu_P)) \cos(n(\mu_P - \mu_Q)) \\ &\quad - \sin(n(\theta - \mu_P)) \sin(n(\mu_P - \mu_Q)) \end{aligned} \quad (28)$$

Taking the expectation and substituting the prepared moments $E_P[\cos(n(\theta - \mu_P))] = \frac{I_n(\kappa)}{I_0(\kappa)}$ and $E_P[\sin(n(\theta - \mu_P))] = 0$, we get:

$$\begin{aligned} E_P[\cos(n(\theta - \mu_Q))] &= E_P[\cos(n(\theta - \mu_P))] \cos(n(\mu_P - \mu_Q)) \\ &\quad - E_P[\sin(n(\theta - \mu_P))] \sin(n(\mu_P - \mu_Q)) \\ &= \frac{I_n(\kappa)}{I_0(\kappa)} \cos(n(\mu_P - \mu_Q)) \end{aligned} \quad (29)$$

Thus, the expectation of the log-denominator term is:

$$E_P[\log(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))] = -2 \sum_{n=1}^{\infty} \frac{\rho^n I_n(\kappa)}{n I_0(\kappa)} \cos(n(\mu_P - \mu_Q)) \quad (30)$$

Substituting this back, we obtain:

$$E_P[\log q(\theta)] = \log(1 - \rho^2) - \log(2\pi) + 2 \sum_{n=1}^{\infty} \frac{\rho^n I_n(\kappa)}{n I_0(\kappa)} \cos(n(\mu_P - \mu_Q)) \quad (31)$$

Combining the results, the KL divergence is:

$$\begin{aligned} D_{\text{KL}}(P \parallel Q) &= E_P[\log p(\theta)] - E_P[\log q(\theta)] \\ &= \kappa \frac{I_1(\kappa)}{I_0(\kappa)} - \log I_0(\kappa) - \log(1 - \rho^2) \\ &\quad - 2 \sum_{n=1}^{\infty} \frac{\rho^n I_n(\kappa)}{n I_0(\kappa)} \cos(n(\mu_P - \mu_Q)) \end{aligned} \quad (32)$$

3.4 Divergence from Wrapped Cauchy to von Mises

The KL divergence $D_{\text{KL}}(Q \parallel P)$ from Q to P is:

$$D_{\text{KL}}(Q \parallel P) = E_Q \left[\log \frac{q(\theta)}{p(\theta)} \right] = E_Q[\log q(\theta)] - E_Q[\log p(\theta)] \quad (33)$$

First, we evaluate $E_Q[\log q(\theta)]$. Using the logarithm of the wrapped Cauchy density:

$$\log q(\theta) = \log(1 - \rho^2) - \log(2\pi) - \log(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q)) \quad (34)$$

Using the Fourier series expansion of the logarithm with $x = \theta - \mu_Q$:

$$\begin{aligned} E_Q[\log(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))] &= -2 \sum_{n=1}^{\infty} \frac{\rho^n}{n} E_Q[\cos(n(\theta - \mu_Q))] \\ &= -2 \sum_{n=1}^{\infty} \frac{\rho^{2n}}{n} \end{aligned} \quad (35)$$

where we have substituted the prepared moment $E_Q[\cos(n(\theta - \mu_Q))] = \rho^n$. Using the Taylor series expansion $-\log(1 - y) = \sum_{n=1}^{\infty} \frac{y^n}{n}$ with $y = \rho^2$, we obtain:

$$E_Q[\log(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))] = 2 \log(1 - \rho^2) \quad (36)$$

Thus, the expectation of the log-density of Q with respect to itself simplifies to:

$$\begin{aligned} E_Q[\log q(\theta)] &= \log(1 - \rho^2) - \log(2\pi) - 2 \log(1 - \rho^2) \\ &= -\log(1 - \rho^2) - \log(2\pi) \end{aligned} \quad (37)$$

Next, we evaluate $E_Q[\log p(\theta)]$. Taking the logarithm of the von Mises density:

$$\log p(\theta) = -\log(2\pi) - \log I_0(\kappa) + \kappa \cos(\theta - \mu_P) \quad (38)$$

To calculate $E_Q[\cos(\theta - \mu_P)]$, we decompose the angle as $\theta - \mu_P = (\theta - \mu_Q) + (\mu_Q - \mu_P)$ and use the addition formula:

$$\begin{aligned} \cos(\theta - \mu_P) &= \cos(\theta - \mu_Q) \cos(\mu_Q - \mu_P) \\ &\quad - \sin(\theta - \mu_Q) \sin(\mu_Q - \mu_P) \end{aligned} \quad (39)$$

Taking the expectation and substituting the prepared moments $E_Q[\cos(\theta - \mu_Q)] = \rho$ and $E_Q[\sin(\theta - \mu_Q)] = 0$, we find:

$$\begin{aligned} E_Q[\cos(\theta - \mu_P)] &= E_Q[\cos(\theta - \mu_Q)] \cos(\mu_Q - \mu_P) \\ &\quad - E_Q[\sin(\theta - \mu_Q)] \sin(\mu_Q - \mu_P) \\ &= \rho \cos(\mu_Q - \mu_P) \end{aligned} \quad (40)$$

Therefore:

$$E_Q[\log p(\theta)] = -\log(2\pi) - \log I_0(\kappa) + \kappa\rho \cos(\mu_Q - \mu_P) \quad (41)$$

Combining the results, the KL divergence is:

$$\begin{aligned} D_{\text{KL}}(Q \parallel P) &= E_Q[\log q(\theta)] - E_Q[\log p(\theta)] \\ &= [-\log(1 - \rho^2) - \log(2\pi)] \\ &\quad - [-\log(2\pi) - \log I_0(\kappa) + \kappa\rho \cos(\mu_Q - \mu_P)] \\ &= \log I_0(\kappa) - \log(1 - \rho^2) - \kappa\rho \cos(\mu_Q - \mu_P) \end{aligned} \quad (42)$$

4 1-Wasserstein Distance (Aligned Means)

When the mean directions are aligned ($\mu_P = \mu_Q$) and one distribution is more concentrated than the other, the 1-Wasserstein distance $W_1(P, Q)$ on the circle reduces to the difference of their mean absolute deviations on $[-\pi, \pi]$:

$$W_1(P, Q) = |E_P[|\theta|] - E_Q[|\theta|]| \quad (43)$$

4.1 Derivation and Proof

Without loss of generality, assume the aligned mean direction is $\mu_P = \mu_Q = 0$ and the distributions are defined on $[-\pi, \pi]$. Since both the von Mises and wrapped Cauchy distributions are symmetric around $\theta = 0$, their cumulative distribution functions (CDFs) satisfy $F_P(-\theta) = 1 - F_P(\theta)$ and $F_Q(-\theta) = 1 - F_Q(\theta)$ for $\theta \in [0, \pi]$.

The 1-Wasserstein distance on the circle S^1 is given by:

$$W_1(P, Q) = \min_{\alpha \in \mathbb{R}} \int_{-\pi}^{\pi} |F_P(\theta) - F_Q(\theta) - \alpha| d\theta \quad (44)$$

Let $D(\theta) = F_P(\theta) - F_Q(\theta)$. By the symmetry of the CDFs, we have:

$$D(-\theta) = (1 - F_P(\theta)) - (1 - F_Q(\theta)) = -D(\theta) \quad (45)$$

which implies that the difference function $D(\theta)$ is odd. The optimal shift α^* that minimizes the integral is the median of $D(\theta)$ over $[-\pi, \pi]$. Since $D(\theta)$ is a continuous odd function on the symmetric interval $[-\pi, \pi]$, its median is exactly 0. Therefore, $\alpha^* = 0$, and the distance simplifies to:

$$W_1(P, Q) = \int_{-\pi}^{\pi} |F_P(\theta) - F_Q(\theta)| d\theta \quad (46)$$

Since $|D(\theta)|$ is an even function, we can restrict the integration to $[0, \pi]$:

$$W_1(P, Q) = 2 \int_0^{\pi} |F_P(\theta) - F_Q(\theta)| d\theta \quad (47)$$

Now, we examine the condition under which the CDF difference $D(\theta) = F_P(\theta) - F_Q(\theta)$ does not change sign (i.e., has 0 crossings) on the open interval $(0, \pi)$. For symmetric, unimodal circular distributions centered at 0, the derivative $D'(\theta) = p(\theta) - q(\theta)$ can change sign at most twice in $(0, \pi)$. Since $D(0) = D(\pi) = 0$, the derivative must change sign at least once.

A necessary and sufficient condition for $D(\theta)$ to have no crossings in $(0, \pi)$ is:

$$(p(0) - q(0))(p(\pi) - q(\pi)) \leq 0 \quad (48)$$

Indeed:

- If $(p(0) - q(0))(p(\pi) - q(\pi)) \leq 0$, the derivative $D'(\theta)$ changes sign exactly once in $(0, \pi)$. Thus, $D(\theta)$ is unimodal and remains strictly positive (if $p(0) > q(0)$) or strictly negative (if $p(0) < q(0)$) on $(0, \pi)$, meaning it has 0 crossings.
- If $(p(0) - q(0))(p(\pi) - q(\pi)) > 0$, the derivative $D'(\theta)$ changes sign exactly twice in $(0, \pi)$. By Rolle's theorem and the Intermediate Value Theorem, this implies that $D(\theta)$ must cross 0 exactly once in $(0, \pi)$.

Therefore, if the condition $(p(0) - q(0))(p(\pi) - q(\pi)) \leq 0$ is satisfied, $F_P(\theta) - F_Q(\theta)$ does not change sign on $(0, \pi)$. Assuming without loss of generality that $F_P(\theta) \geq F_Q(\theta)$ for all $\theta \in [0, \pi]$, we can remove the absolute value:

$$W_1(P, Q) = 2 \int_0^\pi (F_P(\theta) - F_Q(\theta)) d\theta \quad (49)$$

Next, we establish the relationship between this integral and the mean absolute deviation (MAD). The MAD of a symmetric distribution P on $[-\pi, \pi]$ is defined as:

$$E_P[|\theta|] = \int_{-\pi}^\pi |\theta| p(\theta) d\theta = 2 \int_0^\pi \theta p(\theta) d\theta \quad (50)$$

Using integration by parts with $u = \theta$ and $dv = p(\theta) d\theta$ (where $v = F_P(\theta) - 1$), we obtain:

$$\begin{aligned} 2 \int_0^\pi \theta p(\theta) d\theta &= 2 [\theta(F_P(\theta) - 1)]_0^\pi - 2 \int_0^\pi (F_P(\theta) - 1) d\theta \\ &= 2\pi(1 - 1) - 0 + 2 \int_0^\pi (1 - F_P(\theta)) d\theta \\ &= 2 \int_0^\pi (1 - F_P(\theta)) d\theta \end{aligned} \quad (51)$$

Thus, we have:

$$E_P[|\theta|] = 2 \int_0^\pi (1 - F_P(\theta)) d\theta, \quad E_Q[|\theta|] = 2 \int_0^\pi (1 - F_Q(\theta)) d\theta \quad (52)$$

Substituting these expressions back into the simplified 1-Wasserstein integral yields:

$$\begin{aligned}
W_1(P, Q) &= 2 \int_0^\pi (F_P(\theta) - F_Q(\theta)) d\theta \\
&= 2 \int_0^\pi (1 - F_Q(\theta)) d\theta - 2 \int_0^\pi (1 - F_P(\theta)) d\theta \\
&= E_Q[|\theta|] - E_P[|\theta|]
\end{aligned} \tag{53}$$

Taking the absolute value to handle both cases where either P or Q is more concentrated, we arrive at the desired result:

$$W_1(P, Q) = |E_P[|\theta|] - E_Q[|\theta|]| \tag{54}$$

4.2 Analytical Expressions for Symmetric Distributions

The mean absolute deviation of each symmetric distribution is given by:

$$E_P[|\theta|] = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{I_{2k+1}(\kappa)}{(2k+1)^2 I_0(\kappa)} \tag{55}$$

$$E_Q[|\theta|] = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\rho^{2k+1}}{(2k+1)^2} \tag{56}$$

Thus, the 1-Wasserstein distance is:

$$W_1(P, Q) = \frac{4}{\pi} \left| \sum_{k=0}^{\infty} \frac{\rho^{2k+1} - \frac{I_{2k+1}(\kappa)}{I_0(\kappa)}}{(2k+1)^2} \right| \tag{57}$$